

Huntsville, AL

Jorge Mares Zamora
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EDUCATION

University of Alabama in Huntsville (UAH), Huntsville, AL Expected Graduation: Spring 2027

- MS in Aerospace Engineering, GPA: 4.00/4.00

University of Alabama in Huntsville (UAH), Huntsville, AL May 2025

- BS in Aerospace Engineering with a minor in Mathematics, GPA: 3.93/4.00

Select Courses: Computational Fluid Dynamics, Analysis of Engineering Systems, Numerical Analysis, Compressible Aerodynamics, Thermodynamics, Partial Differential Equations, Mechanics of Materials, Aircraft Stability and Control, Astrodynamics.

TECHNICAL SKILLS

Programming Languages: Python, MATLAB, C++, Java.

Other Skills: Fluent in Spanish, 3D Printing and Scanning in Professional Applications, 3D Modeling in Fusion 360 and Solid Edge, Graphic Design in Adobe Creative Cloud and Affinity Software, Photography, Computational Fluid Dynamics, Computing using Linux, Machining with the use of a Mill and Lathe.

EXPERIENCE

Software Developer, GeoRobotix, Inc. (Summer 2025)

- Developed and implemented Java-based sensor drivers and processing pipelines for OpenSensorHub nodes.
- Analyzed and updated technical documentation across the OpenSensorHub codebase to improve clarity, correctness, and maintainability for developers and users.
- Extended existing modules within a large open-source codebase maintaining compatibility with established architectures.

Graduate Teaching Assistant, UAH College of Engineering (Fall 2025 – Current)

- Taught lecture-based numerical methods laboratory sections of 40 students, emphasizing the application of numerical algorithms using MATLAB, alongside discussion of their theoretical foundations.
- Explained and demonstrated numerical techniques such as root-finding, numerical integration, optimization, and solution to ordinary differential equations.
- Debugged student code during lab sessions, reinforcing best practices in numerical accuracy and stability.

Undergraduate Teaching Assistant, UAH College of Engineering (Fall 2022 – Spring 2025)

- Courses: Compressible Aerodynamics and Intro to Computing for Engineers.
- Assisted students with debugging and validating MATLAB and Python solutions to engineering computation and high-speed aerodynamic problems.

RESEARCH EXPERIENCE AND ENGINEERING PROJECTS

Study of Underexpanded Free Jets Produced by Canned Air Dusters.

- Experimental and computational study of underexpanded free jets produced by canned air dusters, combining Schlieren imaging with CFD simulations using OpenFOAM.

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- Designed and tested multiple nozzle geometries to produce and analyze subsonic, underexpanded, highly underexpanded, and extremely underexpanded jets.
- Performed nearfield jet structure Schlieren analysis of the main shock features, including shock diamonds, barrel shocks, and Mach disks.
- Accepted for presentation at the AIAA SciTech Conference (2026).

Development of a Butterfly Inspired Micro Aerial Vehicle with Figure 8 Wing Motion

- Co-authored paper presented during the AIAA student conference (2025).
- Contributed to the design and experimental evaluation of a butterfly-inspired flapping wing micro aerial vehicle aimed to achieve a figure 8 wing motion.
- Supported experimental testing using Vicon motion tracking cameras and force measurements to analyze wing kinematics and aerodynamic performance.

Other Engineering Projects

- UAH Moon Buggy Team: Designed the rear steering locking mechanism for the UAH 2024 – 2025 Moonbuggy team; vehicle placed 3rd in the 2025 NASA HERC competition.
- MATLAB Partial Differential Equation Solvers: Development of solvers to different partial differential equations using MATLAB. Additional projects found at: <http://www.snubmares.com/>

ACTIVITIES

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| • Phi Kappa Psi, Vice President | • Outreach graphic design and photography for the |
| • Board member of SOAR (Service and Civic Engagement Student Organization at UAH) | UAH College of Engineering. |
| • Rock Climbing | • Society of Hispanic Professional Engineers |

HONORS

- UAH Rising Student Leader of the Year.
- UAH President's List (4.0 GPA in a semester, achieved 6 semesters).
- UAH Dean's List (3.5 GPA in a semester, achieved every semester).
- Phi Kappa Psi Foundation District IV Academic Excellence Scholarship Recipient.
- UAH Homecoming King, representing the UAH Service and Civic engagement student organization.
- UAH ACM Hackathon Best Presenter Award.